

System engineer

PROFILE

Software Engineer with expertise in designing and building scalable system architectures. I guide projects through their entire lifecycle—from initial concept to production deployment—by collecting requirements and transforming them into robust software solutions.

EDUCATION

Computer systems engineer

Instituto Tecnológico de Tepic | 2019 – 2026, Tepic, Nayarit, México

SKILLS

- C# | ASP.NET | PYTHON | DJANGO | FLUTTER | JAVA | SYSTEM DESIGN | DESIGN PATTERNS

LANGUAGES

- SPANISH (NATIVE) | ENGLISH (PROFESSIONAL)

PROFESSIONAL EXPERIENCE

Alianza de camioneros del municipio de Tepic

Full Stack Developer | 2025 – PRESENT

Student Credential System for Public Transport - Full-Stack Architecture & Development

C# | ASP.NET | Flutter | Azure web app

- Led complete system architecture design for digital student credential platform enabling urban transportation access
- **Mobile Development:** Built Flutter-based student application for secure storage and display of digital school credentials, deployed across Android and iOS platforms
- **Backend Architecture:** Engineered ASP.NET backend implementing Mediator pattern using MediatR library, architected for seamless future decomposition into microservices architecture
- **Database Design:** Designed PostgreSQL database schema with TimescaleDB extension for efficient time-series event logging (credential validations, access records)
- Delivered end-to-end solution spanning mobile client, API layer, and data persistence with scalability and maintainability as core design principles

Alianza de camioneros del municipio de Tepic

Full Stack Developer | 2025

Public Transportation Feedback Management System

Python | Django | HTML | CSS | JavaScript | Railway

- Developed a complaint management platform enabling transportation organizations to collect and process passenger feedback for their vehicles and routes
- Implemented multi-tenant architecture using Django and django-tenants library on PostgreSQL

- Ensured complete data isolation and privacy between organizations through schema-based tenant separation

DGETI

Full Stack Mobile Developer | 2025

Real-Time Attendance Tracking System - MVP Deployment

Flutter | TypeScript | Vue.js | Supabase

- Built cross-platform Flutter application for teachers and parents to manage and monitor student attendance
- Developed backend with Supabase (PostgreSQL + authentication features) enabling rapid iteration and prototype validation
- Piloted functional prototype at CBETIS 306 to assess market fit and inform roadmap for full-scale development

Estado Limpio

Full Stack Mobile Developer | 2024 – 2025

Waste Collection Tracking System - InnovaTec National Finalist

Flutter | TypeScript | Vue.js | Python | Firebase

- Competed in InnovaTec innovation competition, advancing to national stage after winning local and regional rounds
- Developed MVP for real-time waste collection truck tracking, enabling citizens to monitor collection schedules and vehicle locations
- Built cross-platform solution using Flutter + Firebase for mobile app and Vue.js for administrative dashboard
- Presented prototype to Nayarit State Government, receiving strong interest for potential implementation

Instituto Tecnológico de Tepic

Python Developer | 2024

Python Programming Workshop - Guest Instructor

- Conducted Python fundamentals and applied programming workshop at Instituto Tecnológico de Tepic during Academic Week
- Instructed students from 3rd to 7th semester on language capabilities, automation scripting, and data analysis techniques
- Provided hands-on training in structured data analysis using Pandas library with practical examples

Modelorama CompaBeto

Desktop Developer | 2023 – 2024

Full-Featured Point of Sale System - Custom Development

C# | .Net Window Form

- Engineered bespoke POS solution for Modelorama beer retail client after commercial alternatives failed to meet business requirements
- Delivered integrated system including: sales processing, inventory management, receipt printing, shipment tracking, waste/shrinkage control, employee management, and date-range sales/profit reporting

- Developed using .NET Windows Forms framework with SQLite embedded database for standalone operation and data persistence